

To

Editor, Discover Artificial Intelligence

We are pleased to submit the manuscript titled “Explainable Brain Tumor Classification Using a Hybrid CNN Transformer Network with Cross Modal Context Fusion” for consideration of publication in your prestigious journal “Discover Artificial Intelligence”.

Early and accurate diagnosis of brain tumors is crucial for optimizing patient care and clinical decision making. CNNs are well-suited for capturing local textures and structural patterns in medical images, but they may not be good at capturing long-range spatial dependencies. In contrast, Vision Transformers (ViTs) are better suited for capturing global contextual relationships but can miss out on capturing fine-grained local details that are essential for accurate tumor classification.

To overcome these challenges, this research introduces a CNN–Vision Transformer hybrid network for the classification of brain tumors based on MRI and CT images. The proposed model consists of the MobileNetV2 module for feature extraction in the local part of the image and the Vision Transformer, which has been fine-tuned using the DeiT-Tiny model for global contextual information. A cross-modal attention mechanism is developed, to effectively combine features from both branches, so as to learn local and global representations in a complementary way. The effectiveness of the proposed approach is shown through the experimental evaluation on two benchmark datasets. The superiority of the proposed method was further proved by means of statistical significance analysis. To improve the transparency and clinical trustworthiness, we added Grad-CAM, ViT Attention Rollout, and LIME to offer visual explanations of the model predictions.

By combining effectively local and global feature learning, the proposed framework enhances the interpretability of the classification while retaining high performance, contributing to the field of medical image analysis. We believe this work is well aligned with the scope of Discover Artificial Intelligence Journal, as it advances explainable artificial intelligence and deep learning applications for healthcare and medical imaging. The manuscript is intended for regular submission and has not been published previously, and is not under consideration for publication elsewhere.

Thank you for your consideration. We look forward to your response.

Sincerely,

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